

Cut-Flow Locking and Edgewise Morse Indices on Kuramoto Trees

Route-A source-local certificate HCS-C259

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Abstract

For heterogeneous first-harmonic Kuramoto oscillators on any finite positive-weight tree, we classify every phase-locked relative equilibrium. One cut sum per edge gives a necessary-and-sufficient locking test, the exact branch count, and the complete quotient Morse-index distribution. Equality faces are branch mergers with predictable nullity. The result is a source-local mechanics theorem, not a classification of unlocked motion or an arithmetic determinant.

1 Frozen model

Root a connected tree $T = (V, E)$, $V = \{0, \dots, N-1\}$, at 0 and orient each edge from parent to child. The incidence column has -1 at the parent and $+1$ at the child. For $K = \text{diag}(K_e)$ with $K_e > 0$, the model is

$$\dot{\theta} = \omega - BK \sin(B^\top \theta). \quad (1)$$

The sine is componentwise. Put $\Omega = N^{-1} \mathbf{1}^\top \omega$, $\eta = \omega - \Omega \mathbf{1}$, and $\delta = B^\top \theta$. States differing by a diagonal rotation are identified. Equation (1) is the heterogeneous network form of the Kuramoto phase model [1]; the theorem below is proved directly.

Deleting an edge $e = (p, c)$ leaves a component S_e containing its child. Define the oriented cut demand

$$F_e = \sum_{i \in S_e} \eta_i. \quad (2)$$

2 Locking and branch theorem

Theorem 1 (Complete tree locking atlas). *A locked relative equilibrium exists if and only if*

$$|F_e| \leq K_e \quad \text{for every } e. \quad (3)$$

Its laboratory-frame frequency is Ω . If all inequalities are strict, there are exactly 2^{N-1} locked states modulo diagonal rotation. If exactly s edges are saturated and none is violated, there are exactly 2^{N-1-s} states. If any edge violates (3), there is no locked state.

Proof. Summing (1) fixes the common frequency as Ω . In the rotating frame a locked state obeys $Bf = \eta$, where $f = K \sin \delta$. The tree incidence matrix has rank $N-1$ and therefore determines f uniquely. Summing Bf over S_e cancels internal columns and leaves $f_e = F_e$, proving (2). Thus each edge reduces to

$$K_e \sin \delta_e = F_e. \quad (4)$$

It has two solutions modulo 2π under strict inequality, one at equality, and none outside. Since a tree has no cycle constraint, arbitrary edge differences reconstruct one phase vector after fixing $\theta_0 = 0$. Multiplication gives the stated counts. $\square$

3 Morse theorem

On a local lift of the torus, set

$$V(\phi) = -\eta^\top \phi - \sum_e K_e \cos((B^\top \phi)_e). \quad (5)$$

This is a local lifted potential; the linear term need not define a single-valued global torus potential. Its Hessian at a locked branch is

$$H = B \text{diag}(K_e \cos \delta_e) B^\top. \quad (6)$$

Theorem 2 (Edgewise quotient inertia). *On the diagonal-rotation quotient, the Morse index of a locked branch equals the number of edges with $\cos \delta_e < 0$, and its nullity equals the number of saturated edges. Hence a strict chamber has one and only one linearly asymptotically stable branch modulo rotation; all other strict branches are unstable. Saturated branches are nonhyperbolic.*

Proof. Let the columns of Q span $\mathbf{1}^\perp$ and put $P = B^\top Q$. The square matrix P is invertible because $\ker B^\top = \text{span}\{\mathbf{1}\}$. Therefore

$$Q^\top H Q = P^\top \text{diag}(K_e \cos \delta_e) P. \quad (7)$$

Sylvester’s law proves the inertia statement. The rotating-frame linearization is $-H$, so quotient stability holds precisely when all edge cosines are positive. $\square$

4 Boundaries and Route A

At saturation the inverse-sine branches merge and (7) gains one null vector per saturated edge; no nonlinear stability conclusion is inferred. For $N = 1$ the quotient is a point. If $K_e = 0$, a nonzero cut demand forbids locking while zero demand disconnects the phase constraint, changing the positive-weight owner. Cyclic graphs add flow freedom and cycle constraints, so the theorem does not transfer by relabeling. No global classification of unlocked running states is claimed.

The arithmetic origin is `none`. The strict evaluator tuple is `(AO_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT)` and the verdict is `ROUTE_A_REJECTED`, with `route_b_invocation_allowed: false`, under `NO_BAD_EULER_OR_ROOT_NUMBER`. No target divisor, Euler factor, root number, automorphy, functional equation, or Hilbert–Pólya operator is claimed.

References

- [1] Y. Kuramoto, “Self-entrainment of a population of coupled non-linear oscillators,” in *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Physics 39 (1975), 420–422. <https://doi.org/10.1007/BFb0013365>.